FIRST COURSE HANDOUT,
ELEMENTARY STOCHASTIC PROCESSES I (MTH212M),
2025-26 EVEN SEMESTER (FIRST HALF)
3 - 1 - 0 - 0 [6 CREDITS]

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1. Pre-requisites and Course Objectives

Pre-requisites: MSO201 Probability and Statistics or MSO205 Introduction to Probability or equivalent

Objectives: This course aims to cover the basics of Markov chains. This is a DC course for BS (Statistics and Data Science) and M.Sc (Statistics) students.

2. Course contents and References

2.1. Course contents.

(Topic 1) Definition and classification of Markov processes

(Topic 2) Definition of Markov Chains of Order r (focus on Order 1), and corresponding Transition Probability Matrices

(Topic 3) Examples - Gambler's Ruin, Random Walk etc.

(Topic 4) Chapman-Kolmogorov equations, Derivation of higher order transition probabilities from 1-step transition probabilities.

(Topic 5) Classification of States:
- Accessible, Absorbing and Communicating states
- Communication as an equivalence relation
- Irreducible Markov Chains
- Recurrent and Transient states (focus on the probability of visiting a state infinitely often, through Borel-Cantelli Lemma)
- Effect of dimension on a Markov chain: compare through Random Walks on 1, 2 and 3 dimensions

(Topic 6) Finite Markov chains, properties of finite irreducible Markov Chains

(Topic 7) Periodic States

(Topic 8) Closed sets, Communicating class properties, absorption in closed sets

(Topic 9) Limiting behaviours: Null and Positive Recurrence, Ergodicity

(Topic 10) Stability of Markov Chains, Limiting distributions, Stationary distributions

Date: January 5, 2026.

2.2. **Reference materials.** References:

(a) Samuel Karlin and Howard M Taylor: A First Course in Stochastic Processes, 2ed, Academic Press, 1975.
(b) Sheldon M Ross: Stochastic Processes, John Wiley and Sons, 1996.
(c) Achim Klenke: Probability Theory, Springer-Verlag, 2008.

Supplementary texts:

(a) Robert B. Ash: Basic Probability Theory, John Wiley and Sons, 1970.
(b) Etienne Pardoux: Markov Processes and Applications, John Wiley and Sons, 2008.

2.3. **Important dates.**

Week #	Dates	Remarks
1	January 5 - 9	Start of the course
2	January 12 - 16	
3	January 19 - 23	
4	January 26 - 30	Holiday: January 26 (M), Drop Deadline: January 31
5	February 2 - 6	
6	February 9 - 13	
7	February 16 - 20	
End Sem Exam	February 22 - March 1	As per schedule announced by DOAA
Make-up Exam	March 14-15	As per schedule announced by DOAA

3. Lecture Schedule and Office Hours

- Lectures: M (L4) W (L4) 15:30-16:45 hrs.
- Tutorials: F (L4) 12:00-13:00 hrs.
- Office Hours: Students should communicate their need to discuss doubts by email and take appointments for the same.

4. Weightages for different components of evaluation (out of 100)

Component Name	Weightage
Take-home Assignments	2x10 = 20
End-semester Examination	60
Quiz	20

- End Semester Examination: The examination shall be conducted as per the schedule announced by DOAA. The syllabus, question pattern and other details shall be announced later on.
- Quiz: There shall be one quiz. Syllabi and other details of the quizzes shall be announced from time to time.
- Take-home Assignments: There shall be two take-home assignments, with designated due dates. One question from the question set in each of the assignments shall be chosen by the instructor for grading.
- Exercises/Practice Problems: Exercises and/or practice problems shall be provided on a weekly basis and students are expected to solve them on their own. There is no need to submit the solutions – some of the problems shall be discussed during the weekly tutorial sessions. However, students are strongly encouraged to discuss amongst themselves and also solve problems from the books mentioned above.

5. Course Policies: Attendance, Grading, Withdrawal

- There is no extra weightage for attendance. However, students are strongly advised to attend all the lectures.
- The policy of relative grading will be followed for awarding final grades. However, if the highest marks is below 92, then A* grade may not be awarded.
- No make-up opportunity shall be provided for a missed take-home assignment or a quiz.
- There will be no proration option for a missed take-home assignment.
- In the case where a student misses a quiz due to a bonefide non-emergency situation, then a request for proration may be considered. However, such a request must be made well in advance and the final decision to allow prorating rests with the instructor.
- For missing a quiz due to an emergent situation, request for proration must be made as soon as possible after the quiz date. In case of a medical emergency, the student must present a letter from the doctor stating that the student was not in a condition to take the quiz. The final decision to allow prorating rests with the instructor.
- Make-up option for the End-semester examination will be as approved by DOAA. Applications for the same need to be supported with proof, such as, a medical certificate or a proof of sanctioned leave.
- Any dishonest practice during examinations or quizzes will be reported to DOAA and appropriate action would be taken to penalize such action.
- Students are allowed to withdraw from the course as per the guidelines set by DOAA.